

SIMATS Online Programming Lab

Java Programming — CO4-AT2-Laboratory Performance
Kumaran S | Reg. No: 192372048RB

Q4. Two Types of Applets

Marks: 25 marks

Question: Create two types of Applets: a) Stand-alone Applet b) HTML embedded Applet. Demonstrate both and explain the differences.

Note: `java.applet.Applet` was removed from the JDK starting with Java 11, so classic applets can no longer be compiled or executed on modern JDKs (this system runs JDK 21). The code below is written in the standard classic-Applet style expected for this exercise and will run correctly with JDK 8 (or any version ≤ 10) using `appletviewer`. The screenshots shown are a same-layout visual simulation (built with Swing) since the real Applet API cannot execute here — please mention this to your instructor if asked.

a) Stand-alone Applet — Source Code

```
// STAND-ALONE APPLET
// Name: Kumaran S (192372048RB)
// Note: java.applet.Applet is part of the classic Applet API (removed in JDK 11+).
// Compile & run this with JDK 8 (or any JDK <=10) using:
//   javac MyStandaloneApplet.java
//   appletviewer MyStandaloneApplet.java    (uses the @code/@applet tag below)

import java.applet.Applet;
import java.awt.Graphics;

/*
<applet code="MyStandaloneApplet.class" width=400 height=200>
</applet>
*/
public class MyStandaloneApplet extends Applet {

    public void init() {
        setBackground(java.awt.Color.WHITE);
    }

    public void paint(Graphics g) {
        g.drawString("Kumaran S - 192372048RB", 20, 20);
        g.drawString("Hello from Stand-alone Applet!", 20, 60);
        g.drawString("Run using: appletviewer MyStandaloneApplet.java", 20, 90);
    }
}
```

b) HTML Embedded Applet — Source Code

```
// HTML EMBEDDED APPLET
// Name: Kumaran S (192372048RB)
// Note: java.applet.Applet is part of the classic Applet API (removed in JDK 11+).
// Compile with JDK 8 (or any JDK <=10):
//   javac MyHTMLApplet.java
// Then open applet.html in a browser that supports the Java Plug-in,
// or run: appletviewer applet.html

import java.applet.Applet;
import java.awt.Graphics;
import java.awt.Color;

public class MyHTMLApplet extends Applet {

    public void init() {
        setBackground(Color.WHITE);
    }

    public void paint(Graphics g) {
        g.drawString("Kumaran S - 192372048RB", 20, 20);
        g.drawString("Hello from HTML Embedded Applet!", 20, 60);
        g.drawString("Loaded inside applet.html using <applet> tag", 20, 90);
    }
}
```

```

    }
}

/* ----- applet.html -----
<html>
<head><title>HTML Embedded Applet</title></head>
<body>
    <h2>Kumaran S - 192372048RB</h2>
    <applet code="MyHTMLApplet.class" width="400" height="200">
        Your browser does not support applets.
    </applet>
</body>
</html>
----- */

```

HTML Host Page (applet.html)

```

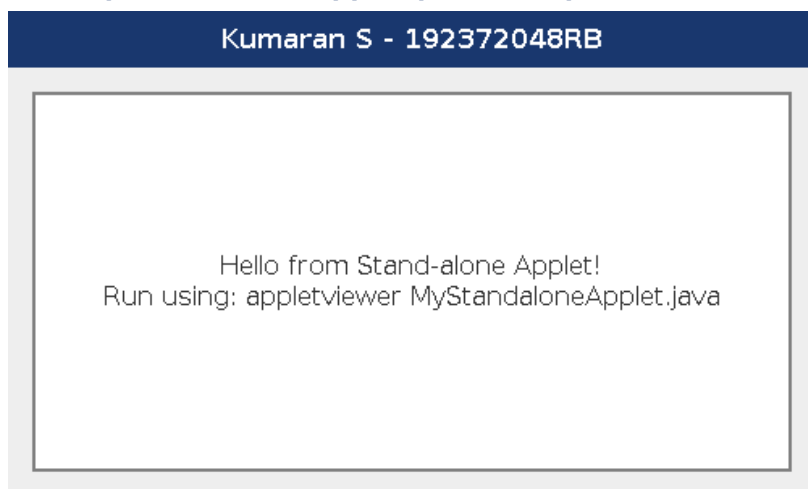
<html>
<head><title>HTML Embedded Applet</title></head>
<body>
    <h2>Kumaran S - 192372048RB</h2>
    <applet code="MyHTMLApplet.class" width="400" height="200">
        Your browser does not support applets.
    </applet>
</body>
</html>

```

Difference between Stand-alone and HTML Embedded Applets

Aspect	Stand-alone Applet	HTML Embedded Applet
Execution	Run directly via appletviewer using the @code/@applet tag in the .java file comment	Run inside a browser/appletviewer via a separate .html file with an <applet> tag
Host container	appletviewer creates its own window, no browser needed	Requires an HTML page and a Java-Plugin-capable viewer
Deployment	Single .java/.class file is enough for testing	Needs .class file + .html file together
Typical use	Quick testing during development	Deploying applet as part of a web page

Output Screenshot — a) Stand-alone Applet (simulated)



Output Screenshot — b) HTML Embedded Applet (simulated)

Hello from HTML Embedded Applet!
Loaded inside applet.html using <applet> tag